

Analysing Methods to Improve Encoder-Decoder Performance for Small Dataset Crop Segmentation

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Abstract—Image segmentation is an important part of smart agriculture. In the last decade, this has been dominated by deep learning, which offers strong performance. However, deep learning approaches often require large datasets, which aren't always available. This paper analyses methods of improving image segmentation for a fully convolutional encoder-decoder network using a small dataset of only 50 images. Concatenation skip connections and data augmentation were the methods used to boost performance. Skip connections bypass the vanishing gradient problem and help the model to avoid training plateaus; data augmentation significantly improves the model's performance on unseen data, due to reducing the effect of overfitting by effectively providing more training data. The improved model reached an accuracy of 92.71%. Despite this performance, the model still has problems — mainly that it produces noisy output and has a tendency to mischaracterise weeds as crop.

Index Terms—Computer Vision, Encoder-Decoder, Crop Segmentation, Data Augmentation, Machine Learning

I. INTRODUCTION

Crop-weed segmentation is becoming a core part of modern smart agriculture, as growers are pivoting away from broadcast spraying with selective herbicides and more towards spot spraying. To be financially viable at a large scale, spot spraying must be automated. The core part of this automation is a segmentation model that is able to segment weeds from crop when fed an input image. Early crop-weed segmentation dates back to the 1990s and early 2000s using hand-crafted computer vision techniques and classical computer vision [1], with modern segmentation approaches emerging after 2015, following the introduction of various deep learning architectures [2]. However, deep learning methods typically require large amounts of training data in order to achieve good performance — a restriction that often limits their effectiveness. It is not always feasible to obtain large amounts of training data, so this presents a significant challenge for applying deep learning methods to agricultural environments. This is particularly relevant in crop-weed segmentation, where labelled datasets are often limited due to the cost and time required for manual annotation.

This paper explores techniques to bypass the problems associated with small datasets for deep learning methods of crop-weed segmentation. An encoder-decoder model [3, 4] was used as a baseline and subsequently improved using data augmentation and skip connections. The performance

of the baseline and improved models were evaluated using mean accuracy, mean intersection-over-union (IoU) and mean Boundary F1 (BF) score. Figure 1 shows typical outputs of the baseline and improved models.

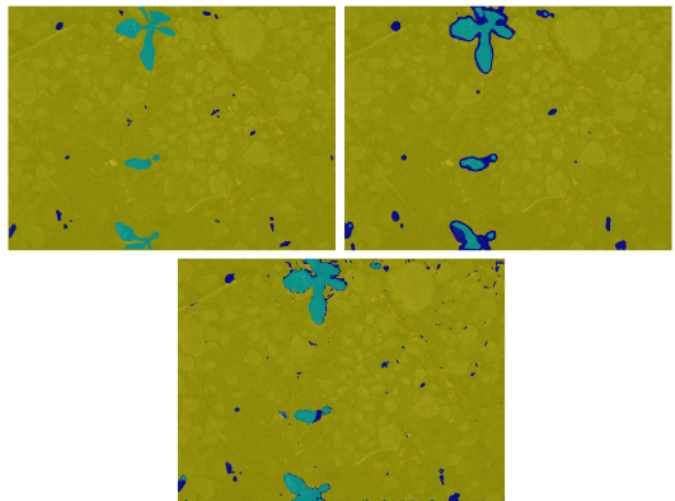


Fig. 1. **Example model outputs.** The baseline (top-right) and improved (bottom) model outputs compared with the ground-truth label (top-left).

II. METHOD

To analyse the impact of different improvement methods, four models were created. Apart from each having a different combination of improvements implemented, these models were identical and created using the same method, as described below.

A. Data Pre-processing

The dataset of fifty image-pair labels used for the training of the model initially stored the ground-truth labels as RGB images, a format incompatible with the model. The RGB images were converted into a single-channel format, with the value of each pixel representing its class.

The inverse class frequency was then calculated from the training data to balance class weights during training. Without this metric, the model collapsed into exclusively predicting the majority class (the ground class), because it made up 95% of the training data, as shown in Table I.

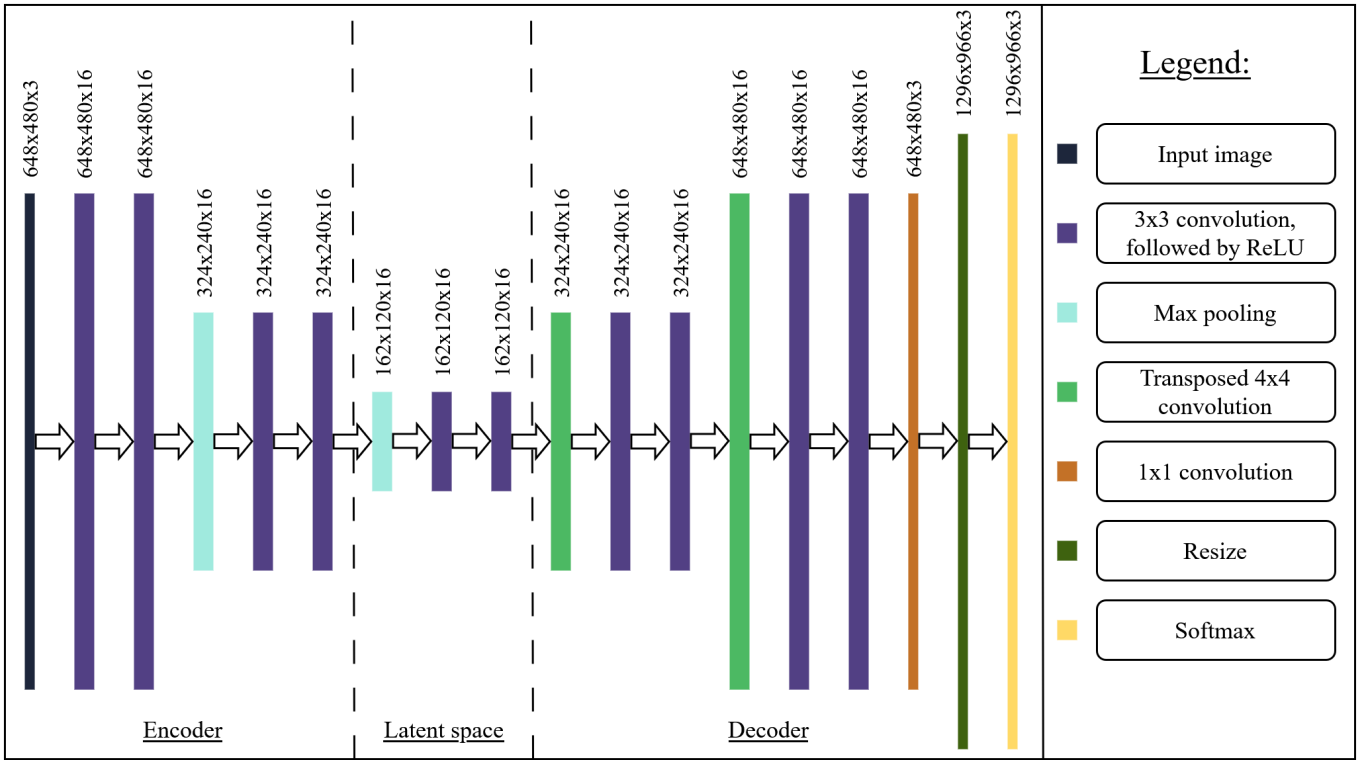


Fig. 2. **Network architecture.** The architecture used by the models (left) and the corresponding legend (right). The encoder is shown on the left side of the network, the latent feature space in the center, and the decoder on the right. The resolutions of the input for each layer of the network are shown above the layer's representation.

TABLE I
CLASS FREQUENCY AND INVERSE FREQUENCY

Class	Ground	Crop	Weed
Frequency	0.95	0.04	0.01
Inverse Freq.	1.06	26.08	61.43

An 80/20 training/test ratio was used to partition the data. The data was partitioned randomly to prevent the ordering of the dataset impacting the model. Finally, input images were resized from 1296×966 to 648×480 before being used for training. This was done to reduce the computational load of processing the images, thus reducing the training time required by the model. The ground-truth labels were unchanged to preserve information that would otherwise be lost through resizing.

B. Model Architecture

The baseline model consisted of two sub-structures, an encoder and a decoder, as shown in Figure 2. This architecture was chosen as it has historically performed well for image segmentation tasks [4]. The encoder was responsible for taking an input image and compressing it to a latent feature space. The decoder then took latent space features as input and reconstructed them into an image.

1) *Encoder:* The encoder was made of three layers (including the latent space), each with different resolutions. At the starting layer, the encoder took a 648×480 RGB image as input. The encoder used 3×3 convolutions, followed by a ReLU [5] activation layer to facilitate non-linear learning. ReLU was chosen as the activation function instead of tanh (a commonly used alternative) because ReLU has been proven to be much faster. For each layer, the convolution-ReLU process occurs twice before the image is downsampled using a max pooling layer.

2) *Decoder:* Like the encoder, the decoder was also made of three layers. The first and second decoder layers matched the resolution of the second and first encoder layers, while the third decoder layer resized the images to 1296×966 . The decoder used the same convolution-ReLU units as the encoder, and used a transposed 4×4 convolution to upsample the image. A 1×1 convolution in the second layer was used to reduce the number of channels before resizing. After the resize, a softmax layer was used to convert scores into class probabilities.

C. Model Training

The model was trained for thirty epochs with an initial learning rate of 0.001. ADAM [6] was selected as the optimiser to be used during training due to its strong performance when training convolutional neural networks. To smooth the learning signal obtained from the training data, images were fed to the

network in a mini-batch of eight. Lastly, it was crucial that the dataset was shuffled after each epoch. This reduced the effect of anomalous data on the learning signal obtained from each mini-batch. Following the 80/20 split, each model had forty images (before data augmentation) available during training.

D. Model Improvements

To improve upon the initial model, skip connections and data augmentation were used. Different combinations of these methods yielded performance improvements over the baseline model.

1) *Skip Connections*: Concatenation skip connections work by concatenating the output of one layer to the input of another layer later in the network. These were used in the model to decrease the effects of the vanishing gradient problem [7], whilst also preserving detail lost during downsampling. Skip connections were chosen for this network to convert the encoder-decoder to a U-net architecture [8], which have consistently been shown to perform well for image segmentation problems.

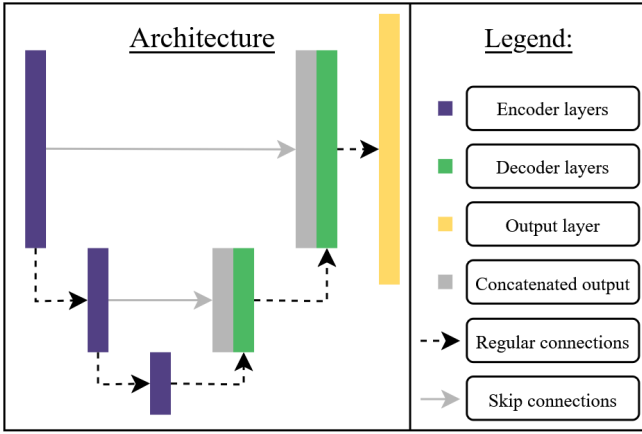


Fig. 3. **Skip connections.** Simplified network architecture including concatenation skip connections (left) and the corresponding legend (right). See Figure 2 for a further example of model architecture.

2) *Data Augmentation*: Augmenting the original dataset helped to reduce the effect of overfitting and improved the model’s ability to generalise. To achieve this, data augmentation was applied during training for each image fed into the network. The augmentations were applied randomly, and included: flipping the image along the X-axis, flipping the image along the Y-axis, rotating the image by $\pm 10^\circ$, shearing the image along the X-axis in the range $[-0.1, 0.1]$ and uniformly scaling the image in the range $[0.9, 1.1]$.

It was important that the ground-truth labels were also augmented to remain consistent with the training images. However, the resizing of labels had to be done via a nearest neighbour resizing method instead of bilinear resizing to avoid corrupting class labels.

III. MODEL EVALUATION

In this section, the results of running the different models on unseen data are revealed and interpreted. These results are

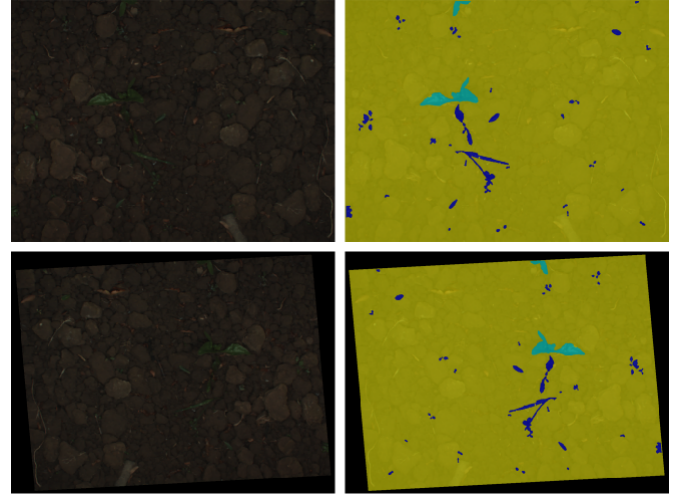


Fig. 4. **Data augmentation.** Example of an image-label pair before (top) and after (bottom) data augmentation. Ground-truth labels are shown on the right: Yellow represents the ground class, cyan the crop class, and blue the weed class.

based on a ten image test set that the model did not see in training.

As shown in Table II, combining both improvement methods yielded the highest mean accuracy for this instance of test data. Notably, skip connections without data augmentation seemed to harm the model’s learning in this instance as, while the baseline and skip connection models have a similar mean accuracy, they differ in mean IoU and mean BF score.

Of the improvement methods, data augmentation had the highest impact on model performance for unseen data.

TABLE II
EVALUATION METRICS OF DIFFERENT SEGMENTATION MODELS

Metric	Mean Acc (%)	Mean IoU (%)	Mean BF (%)
Baseline Model	89.70	66.93	73.30
Augmentation	92.16	61.96	67.64
Skip Connections	89.30	51.83	45.67
Augment + Skip	92.71	64.20	60.92

A. Discussion of Results

1) *Skip Connections*: The discrepancies in mean IoU and mean BF scores between the baseline and skip connection models imply that the skip connections somewhat reduced the model’s ability to identify whole regions of crop and weed effectively, even if the total number of pixels correctly classified by each model were equal. Although the accuracy of the model was reduced, skip connections sped up the training process substantially. The likely reason for this is that the concatenated data from the skip connections helped bypass the vanishing gradient problem, allowing for faster optimisation. This would allow for a potentially more complex and deeper model to be used, while still maintaining a reasonable training speed.

2) *Data Augmentation*: Whilst the accuracy during training for the baseline and augmentation models were similar, there was a large difference in performance when feeding the models unseen data. This was likely due to the baseline model being heavily overfitted to the training data, which is a common problem for training with smaller datasets. Data augmentation increases the variation of a dataset, reducing the model's tendency to overfit.

3) *Model Limitations*: As shown in Figure 5, the output of the improved model had a tendency to be noisy. This likely means that further improvements could be made to the model by either incorporating image blur into the data augmentation or applying blur to the output of the model itself.

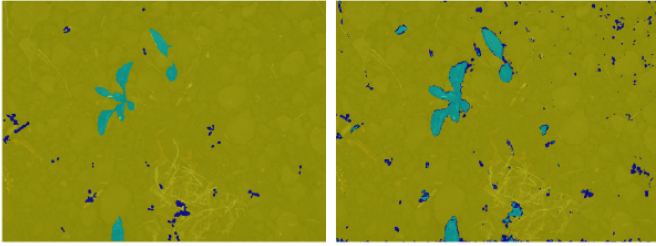


Fig. 5. **Noisy Output.** The ground-truth (left) compared with the improved model's output (right). The output of the improved model was noisy, showing false positive weed predictions in ground regions.

The improved model still occasionally misidentified weeds as crops. This could be improved in future by creating a deeper model with further augmentations to reduce overfitting.

4) *Evaluation Limitations*: The metrics shown in Table II were gathered using a 10 image test set. Due to this small test set, the evaluation metrics have a high variance. To improve upon this, future methods could use k-fold cross-validation to help reduce the variance of the results.

IV. CONCLUSION

In this study, data augmentation was the most effective method for improving generalisation for reduced dataset problems. Whilst other methods exist and help in a limited capacity, augmentation keeps overfitting low and improves the model's ability to generalise. Future models could also utilise regularisation and dropout [9], which have also been shown to consistently reduce overfitting.

This study demonstrates that for relatively shallow models, skip connections appear to have negligible impact, as the benefit of bypassing the vanishing gradient problem seems too small to outweigh the cost of disturbing the learning from the compressed features.

Future research could explore different network architectures to test if the lack of sufficient data more heavily impacts one architecture over another.

Overall, these results demonstrate that data augmentation can improve the viability of image segmentation approaches for practical agricultural applications where labelled data is scarce.

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